Effects of a Nuclear Detonation in Low Earth Orbit

Evident Lab

University of Wisconsin-Madison

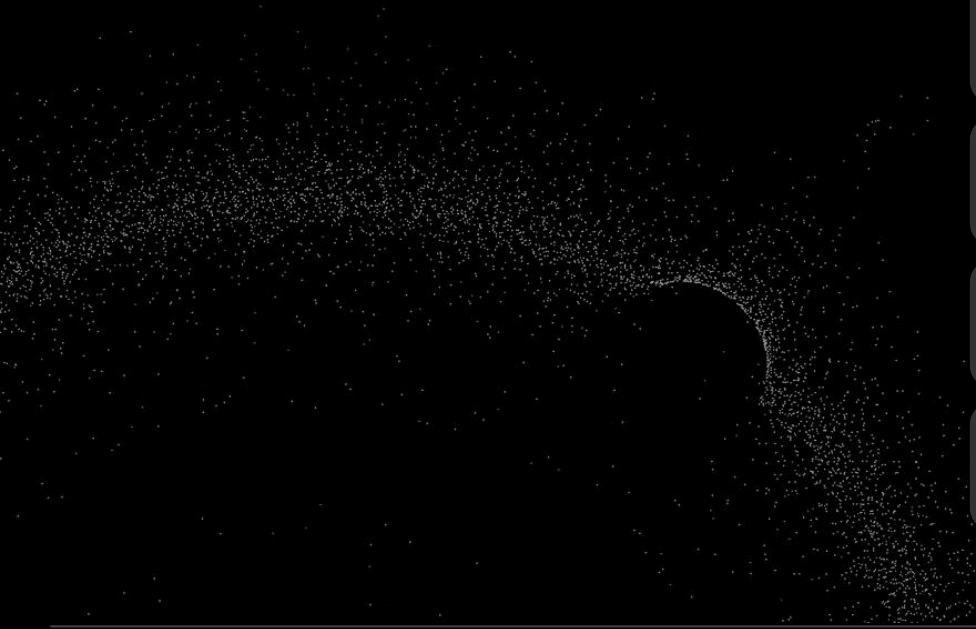
Presented by

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Collaborators

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Agenda



1 Current Space Environment

2 HANE Overview

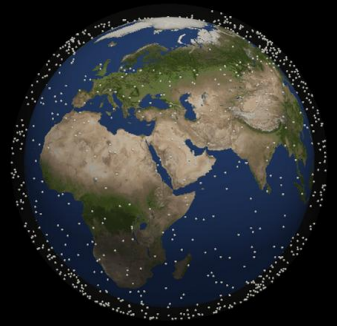
3 Prompt X-ray Effects

4 Belt Trapping Effects

5 EMP

6 Conclusion

Current Space



Mega constellations contain **thousands** of satellites each

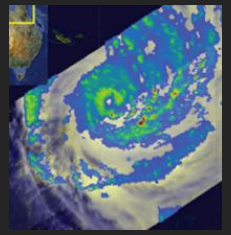
13,305

Satellites in Low Earth Orbit

x10

Increase in total satellites from 2024 to 2030 (European Space Agency)

Uses



Weathe

STARLINK

Communication



Military Reconnaissance



Space Exploration

and much more

Images from NASA, Starlink, Northrop Grumman
Constellation data from Celestrak (2026-01-14)

Nuclear Anti Satellite Weapon

- Desire to damage adversary forces/infrastructure
    - NC3 systems in orbit and global economic effects
- Possible use as a nuclear warning shot
- Cause damage with lower likelihood of nuclear retaliation
    - Can be used to disrupt electronics from afar
    - Avoids direct lethal effects

Outer Space Treaty: nuclear weapons shall not be stationed in orbit or elsewhere in outer space



February 2022 Russian satellite Cosmos 2553 was sent into LEO
Photograph: Shamil Zhumatov/Reuters

Overview of HANE Effects

Prompt X-rays

Immediate

Nuclear explosion releases energy as x-rays which travel through the atmosphere and destroy nearby satellites

Electromagnetic Pulse

Immediate

Gamma pulse ionize air and release electromagnetic energy. Creates blackouts on the ground

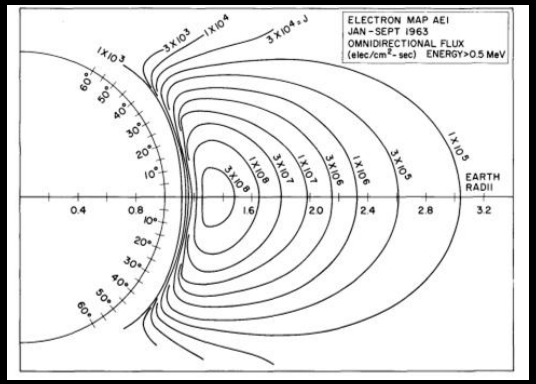
Artificial Belt Pumping

Delayed

Fission debris decay creating electrons that become trapped in

The peak radiation flux increases by a factor of 10^3 to 10^4

DTRA 2001 HALEOS study



Post Starfish Prime Flux from The Trapped Radiation Handbook



Starfish Prime, 1962, 1.4Mt @ 400km
Visible aurora from Honolulu
8/24 satellites in orbit destroyed

Research Question

Could a high altitude nuclear explosion quickly kill all LEO?

NASA investigates high energy charged particle radiation environments that cause charging on satellite solar panels

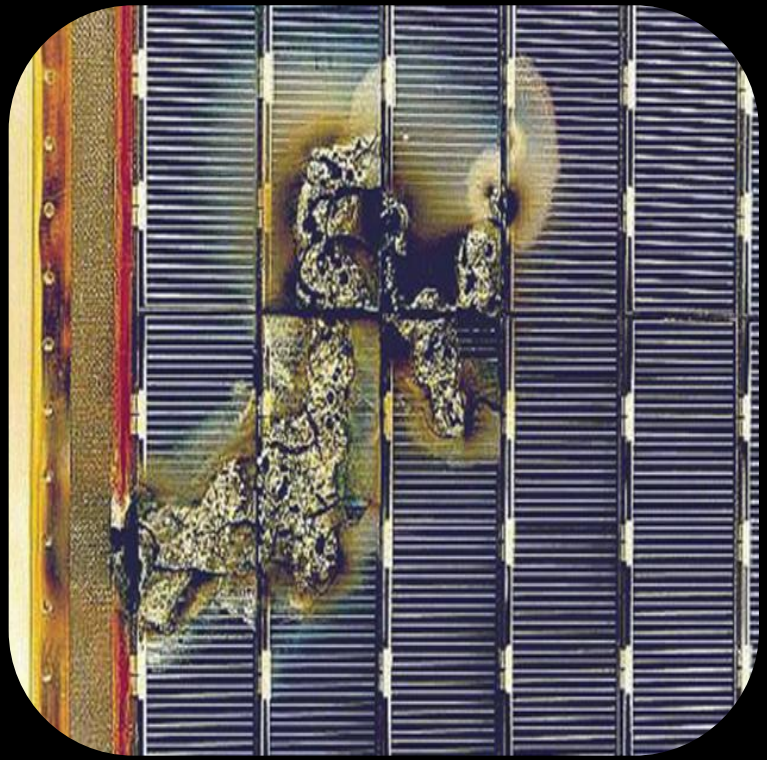


Image from NASA

High Altitude Nuclear Explosions Create Prompt Gamma Rays

- X-rays account for 70-80% of the weapon yield

- High radiation instantly kills satellite function

$$\text{Fluence} = \frac{\text{yield} * \text{Xray Fraction} * 4.184\text{E}12\ [\text{J/kT}]}{4\pi r^2}$$

Scenario

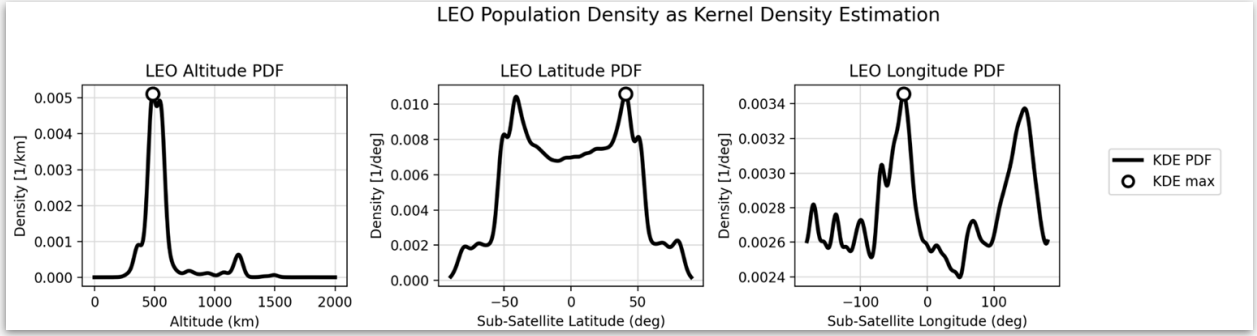
- 1 MT
- Burst height of 485 km
- Above the Atlantic Ocean
- Satellites withstand 40 J/m^2

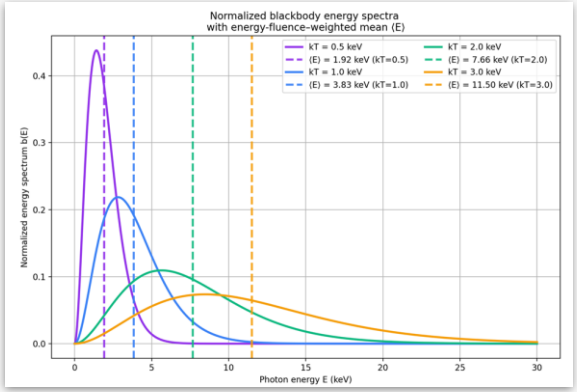
Damage

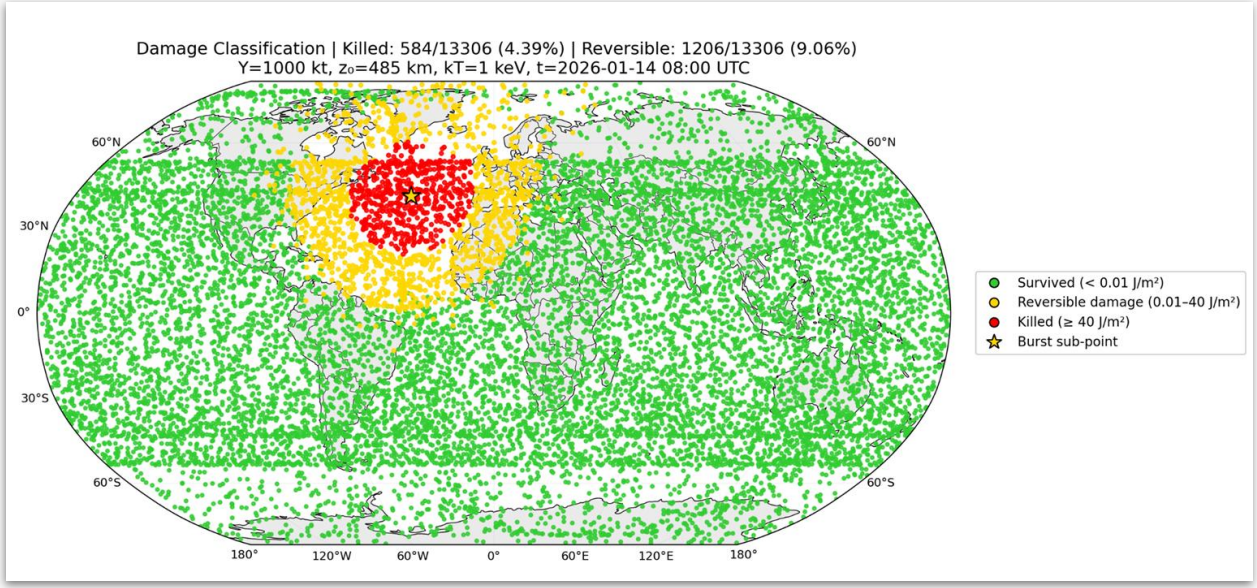
- Killed **4.39%** of all LEO satellites
- ~1800 will be destroyed or temporarily inoperable
- Indiscriminate against country of origin

*Scenario fluence includes blackbody energy spectrum folding and attenuation in atmosphere

Math from Fang Zhengqing. Assessing the Effect of High-Altitude Nuclear Explosions (HANE) against LEO Satellites.



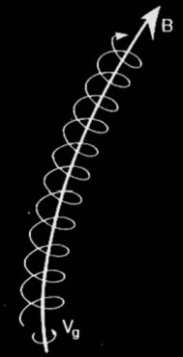




Destroyed **4.39%** of all LEO satellites from prompt radiation

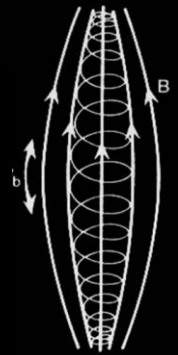
Creation of the artificial radiation belt

1



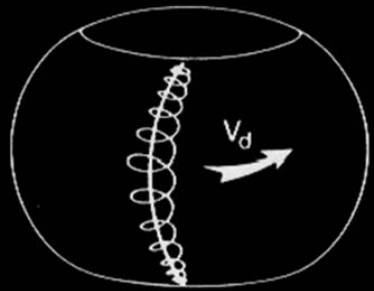
Gyro
Motion around field lines
Period of 0.1 ms

2



Bounce
Motion along field lines
Period of 0.36 s

3



Drift
Motion around axis of symmetry
Period of 15 min

Images from Geomagsphere.org
Periods from LANL ISR-1 Greg Cunningham

Trapping Mechanics

Momentum p and Lorentz factor γ

$$p = \gamma m v, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \qquad p_\perp = p \sin \alpha_{eq}, \qquad p_\parallel = p \cos \alpha_{eq}$$

Equatorial pitch angle α_eq and mirror condition

$$\sin^2 \alpha_{eq} = \frac{B_{eq}}{B_m}$$

First adiabatic invariant μ (magnetic moment)

$$\mu = \frac{p_\perp^2}{2mB}$$

Second adiabatic invariant K (bounce invariant)

$$K = \int_{s_1}^{s_2} p_\parallel \, ds$$

Losses

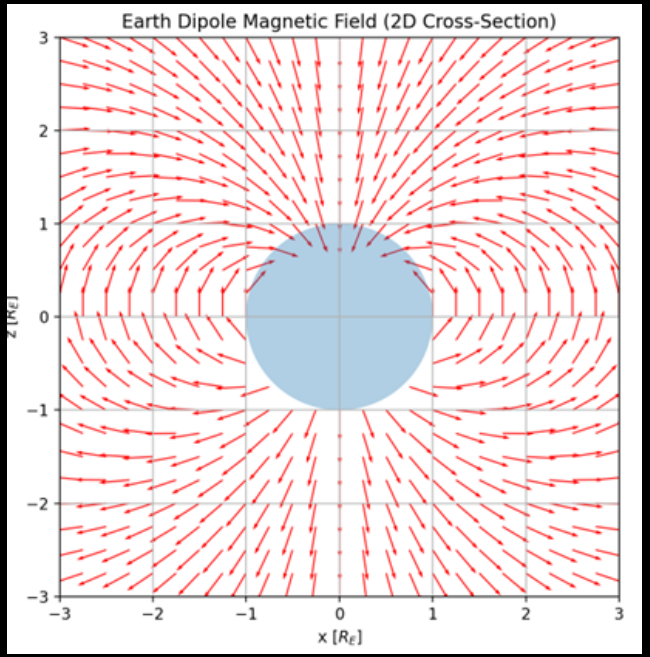
If particles encounter atmosphere before bounce they will be lost by absorption

$$for\ \alpha < \alpha_{LC} \quad \text{where } \sin^2 \alpha = \frac{B_X}{B_a}$$

x is any location

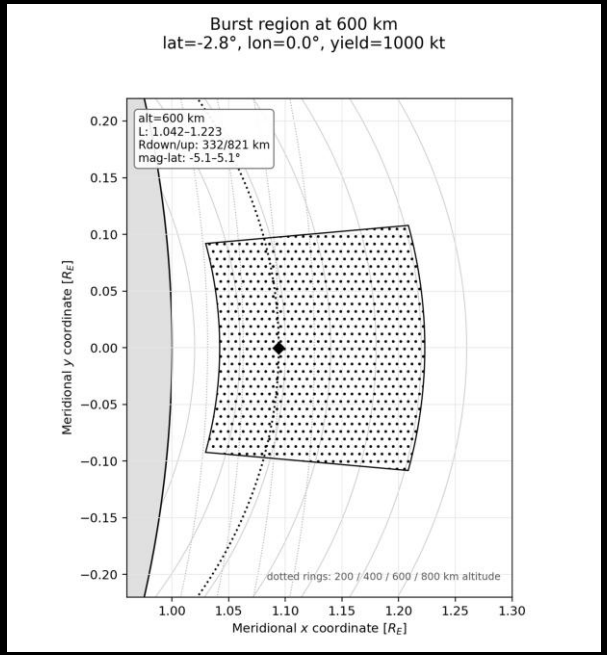
a is intersection of field line with atmosphere

Model Verifications



Create Magnetic Field

Use a simple dipole approximation



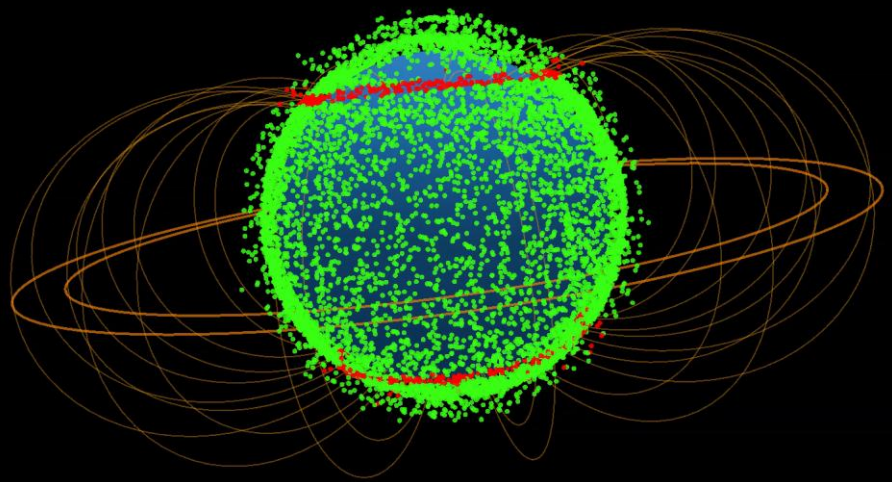
Calculate Burst Region

Use "equal mass radius" and "magnetic confinement radius". Compared with LANL ESM results

Results

Satellites entering the artificial belt (cumulative) — L 2.15–2.40

red = has passed through the trapped region · 920/13306 (7%) hit by frame 0 (t+0 min)

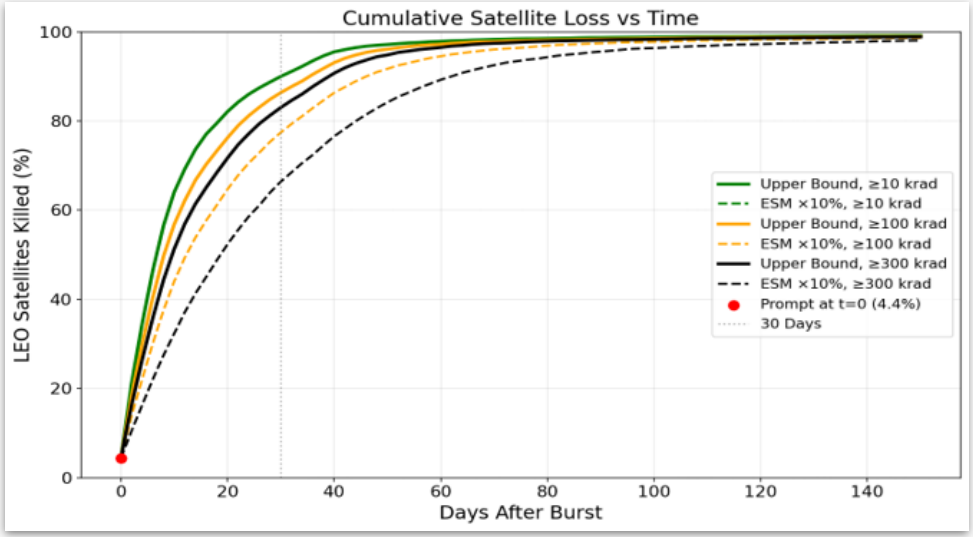


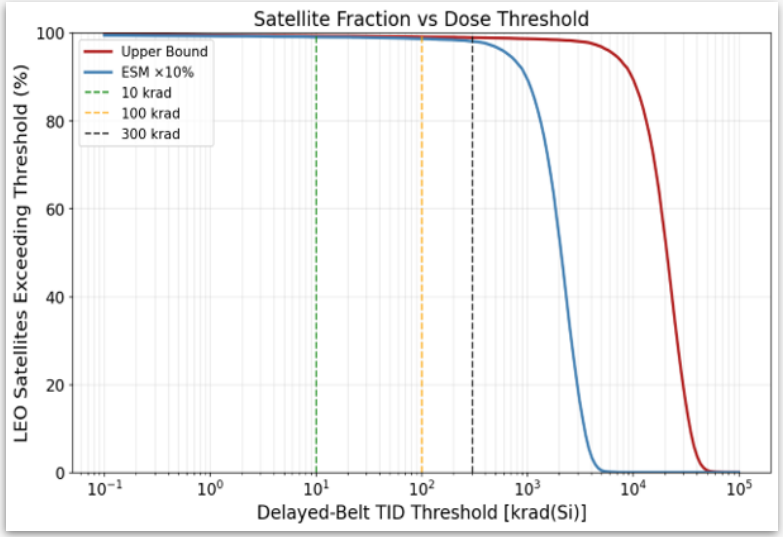
The trapped electrons drift in longitude to populate a complete shell within minutes

Threatens any satellite independent of burst longitude

64% of LEO satellites cross the injected band within the first hour

Results

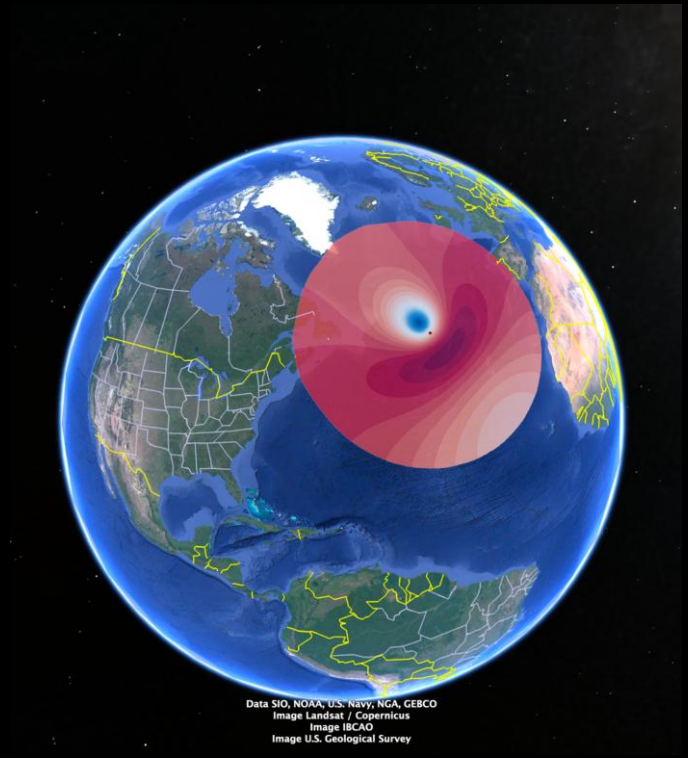




Assuming 2 mm shielding and 10% electron-trapping efficiency 97.5% of satellites exceed 10krad(Si) within 30 days, while 90% exceed 300 krad(Si) within 62 days.

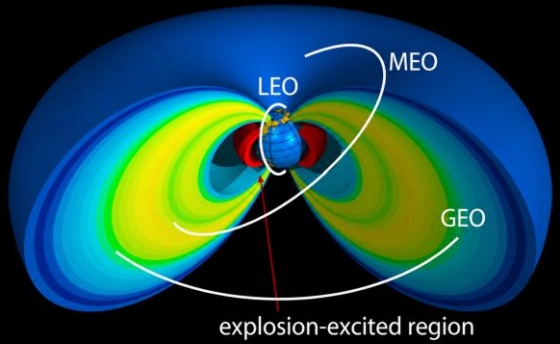
A HANE causes large ground EMP

- E1 from a 1MT, HOB 485 km, lat: 40.75, lon: -35.50
- The colors match a logarithmic scale from dark blue (0V/m) to deep red (135,000V/m)
- Diameter **~4,400km**
- At higher altitudes range increases and strength decreases



The Karzas-Latter-Seiler Model of a High-Altitude Electromagnetic Pulse

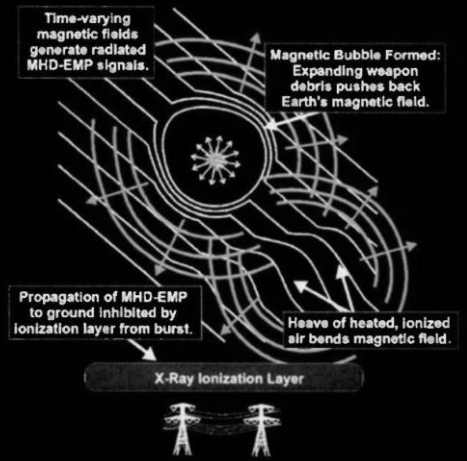
Next Steps



Model improvements

Real magnetic field line tracing, improve expansion and loss terms

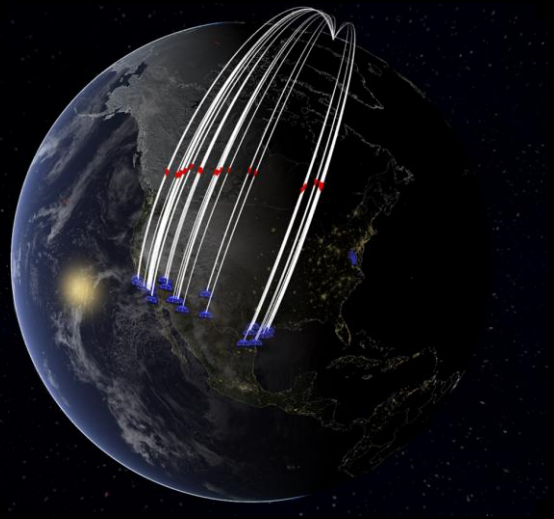
Image from DTRA



Ground effects

Better understanding of the EMP effects to the grid

Image from Collateral Damage to Satellites from an EMP Attack



NWS

Implement HANE effects in Nuclear War Simulator

Conclusion

1

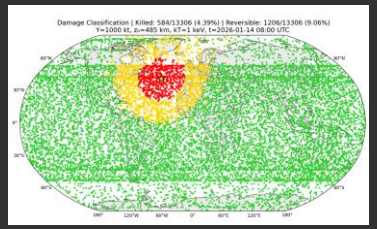
Satellite population is exponentially increasing in size and importance



Northrop Grumman

2

Prompt gamma rays will immediately disable a fraction of LEO satellites



3

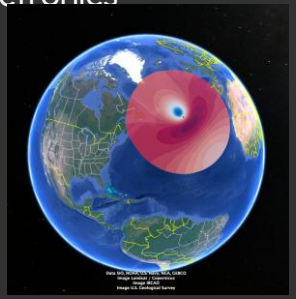
Trapped betas could quickly kill all LEO satellites



NASA

4

Large scale EMP can target telecommunications and electronics



Citations

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Thank you

Contact Me

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